Monthly report (2/16/2024)

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Outline: This report discusses my attempts at simulating droplet spreading on a wall after impact. It is organized as follows: first I discuss why we chose to simulate this problem. Secondly, the computational method and its limitations concerning the droplet impact problem, as shown by several related experiments, are discussed. Thirdly, how I got around some of these limitations by customizing the code is discussed. Lastly, the possible direction of this work is talked about.

1. Introduction

Supercooled water droplets make up a considerable proportion of clouds at higher altitudes. These droplets have not yet frozen, and an impact is all they need to start nucleating. An aircraft often flies at similar altitudes to reduce fuel consumption. Whenever it passes through clouds, the surfaces of the fuselage and the wings hit the suspended liquid droplets at high speeds, freezing them in the process. As a result of many such impacts, eventually a large portion of aircraft surfaces are coated with ice sheets. This alters the aerodynamic shape of these surfaces, affecting the very purpose of flying at such altitudes. The aircraft industry has come up with several ways to mitigate this problem. Thermal, electromechanical deicing systems, ice phobic coatings, advanced fluids for deicing are the most prominent methods. The global aircraft deicing market was $1.2 billion in 2021 and is expected to grow to $1.6 billion in 2027. However, none of these methods can justify their efficiency against the associated costs.

Hence, such icing effects are assessed through extensive procedures under airworthiness agencies certification, such as the Federal Aviation Administration's (FAA) Appendix C [1]. More recently, the FAA introduced Appendix O [2] mandating new aircraft under 60,000 lbs to be protected against the effects of supercooled large droplets (SLDs). SLDs are defined as droplets of a median volume diameter higher than 50 mm, suspended in liquid form at below freezing conditions [2]. Because of the large size, SLDs impingement leads to droplet break-up, splashing, rebound and deposition or spreading [3]. The icing envelopes (Appendices C and O) cannot be fully certified exclusively using either icing-tunnel tests or flight tests. Airworthiness under icing conditions has thus traditionally been demonstrated by a judicious combination of computational fluid dynamics simulations, experimental investigations in dry and ice tunnels and flight testing in natural icing [4]. SLD conditions are, however, particularly difficult to replicate in icing tunnels as droplets may settle at their base before reaching the test section. They are also difficult to locate in nature. Numerical approaches have thus become an indispensable component of the certification process.

Our primary goal is to investigate the impingement of a single SLD, to gain microscopic knowledge which can be incorporated into the macroscopic models currently embedded into the in-flight icing codes. The challenge of numerically simulating SLD impingement arises from the complex interface behavior that requires special modeling techniques. Mesh-based methods use either interface-tracking or interface-capturing to deal with moving interfaces. In interface-tracking, meshes conform to the interfaces and are updated as the flow evolves. They are used for small interface motions but may not be suitable for large deformations as continuous re-meshing is required [5]. In this case, interface-capturing may be more flexible and efficient compared to interface-tracking. Interface-capturing can decrease the computational time significantly for handling complicated interface motions [5]. Numerical experiments on high-speed droplet impingement were carried out [6,7], wherein the interface was captured using the level-set method and the moment-of fluid method, respectively. However, interface-capturing methods cannot strictly satisfy mass conservation [5]. Alternatively, the smoothed particle hydrodynamics (SPH) method can implicitly conserve mass and handle complex interfaces, making it an attractive approach for studying SLD dynamics.

The SPH method is a Lagrangian mesh-free particle method [8,9], which has been used to simulate various fluid phenomena, such as free surface flow [10,11,12], multiphase flow [13, 14, 15], turbulent flow [16, 17, 18], explosion [19,20], fluid–structure interaction [21, 22], drop collisions [23,22], viscoelastic drop impact on a solid surface [24, 25], and drop flow on surfaces [26]. The mesh-free nature of the SPH method reduces the difficulties in tracking free surfaces because SPH uses particles, rather than mesh, as a computational frame to solve governing equations.

In this report, the open-source SPH software SPHinXsys [27] is used to simulate droplet impact on a solid wall. The behavior of a droplet after impact has been experimentally characterized into five types – stick, spread, fingering, splashing and rebound [28,29,30]. These outcome(s) are dictated by properties of the liquid, fluid surrounding the droplet, droplet size and shape, impact velocity and properties of the wall. Since, SLD involves phase change in addition to droplet impact dynamics, we attempt to first reproduce results of numerical experiments on low-speed single droplet impingement by Yang et al. [31].

2. Methodology

2.1 Governing equations

In the Lagrangian frame, the conservation of mass and momentum for fluid dynamics can be written as:

$$\left.\begin{aligned} \frac{\partial \rho}{\partial t} &= -\rho \nabla . \boldsymbol{v} \\ \frac{\rho \partial \boldsymbol{v}}{\partial t} &= -\nabla p + \eta \nabla^2 \boldsymbol{v} + \rho g \end{aligned}\right\} \quad \textbf{(1)}$$

where v is the velocity, ρ the density, p the pressure, η the dynamic viscosity, g the gravity and $\frac{d}{dt} = \frac{\partial}{\partial t} + \boldsymbol{v}.\nabla$ stands for the material derivative. For modeling incompressible flow with weakly compressible assumption [32,33], the WCSPH method uses an artificial isothermal equation of state (EOS) to close Eq. (1)

(2)

$$p = c^2(\rho - \rho_0)$$

Usually, the density varies around 1% [33] if an artificial sound speed of c=10U_{max} is employed, with U_{max} being the maximum anticipated flow speed.

Following Refs. [34,35,36,37], a standard WCSPH discretization of the continuity equation is

$$\frac{d\rho_i}{dt} = 2\rho_i \sum_j \frac{m_j}{\rho_j} (\boldsymbol{v}_i - \bar{\boldsymbol{v}}_{ij}). \nabla_i W_{ij} \qquad \textbf{(3)}$$

Here, m_j is the particle mass, $\bar{\boldsymbol{v}}_{ij} = (\boldsymbol{v}_i + \boldsymbol{v}_j)/2$ is the average velocity between particle i and j. $\nabla_i W_{ij}$ is the gradient of the kernel function W (|$\boldsymbol{r_{ij}}$|,h), where $\boldsymbol{r_{ij}} = \boldsymbol{r_i} - \boldsymbol{r_j}$, and h is the smoothing length with respect to $\boldsymbol{r_i}$, the position of particle i. In this work, the cubic spline kernel is used:

$$W(s,h) = \begin{cases} \sigma_3 \left[1 - \frac{3}{2} s^2 \left(1 - \frac{s}{2}\right)\right], & for\ \ 0 \leq s \leq 1 \\ \frac{\sigma_3}{4} (2-s)^3, & for\ 1 < s \leq 2 \\ 0, & for\ s > 2 \end{cases} \qquad \textbf{(4)}$$

where $\sigma_3 = \frac{10}{7\pi h^2}$, $\sigma_3 = \frac{1}{\pi h^3}$ for two and three dimensions respectively.

A standard discretization of the momentum equation without considering artificial viscosity [36,38] is:

$$\frac{dv_i}{dt} = -2 \sum_j \frac{m_j \bar{P}_{ij}}{\rho_i \rho_j} \nabla_i W_{ij} + \sum_j \frac{m_j (\mu_i + \mu_j) (\boldsymbol{r_i} - \boldsymbol{r_j}). \nabla_i W_{ij}}{\rho_i \rho_j (r_{ij}^2 + \eta_1)} (\boldsymbol{v_i} - \boldsymbol{v_j}) \qquad \textbf{(5)}$$

where $\bar{P}_{ij} = (P_i + P_j)/2$ is the average pressure between particle i and j, μ_i, μ_j are the dynamic viscosities of particles i and j, $\eta_1 = 0.01h^2$ is added to prevent the singularity of the viscous term. In standard WCSPH the discretized momentum equation also includes an artificial viscosity term

$$\Pi_{ij} = -2\alpha ch(v_i - v_j). \frac{(\boldsymbol{r_i} - \boldsymbol{r_j})}{(\rho_i + \rho_j)(r_{ij}^2 + \eta_1)} \ , \quad (\boldsymbol{v_i} - \boldsymbol{v_j}). (\boldsymbol{r_i} - \boldsymbol{r_j}) < 0 \qquad \textbf{(6)}$$

where $\alpha \leq 1$ is a tunable parameter to control the numerical damping. A particle-based method like SPH is prone to unphysical particle oscillations because it discretizes a continuum to discrete particles. The concept of artificial viscosity was introduced in SPH to address this issue [41].

While moderate artificial viscosity can stabilize the computation, for some problems it may lead to excessive dissipation which affects the physical flow characteristics [42, 43]. Another weakness is that the tunable parameter α requires careful numerical calibrations and its values usually are case dependent [44].

SPHinXsys by default uses the Riemann-based SPH method which introduces implicit numerical dissipation and achieves the dissipation in a more accurate manner [42, 45-47]. Monaghan [45] pointed out that the artificial viscosity [38] is analogous to the dissipative terms of the Riemann solver, which scales with the wave speed and the velocity jump between interacting particles, whereas shown that using the exact, or well approximated, Riemann solution [40] can obtain more accurate results in capturing shock waves.

The exact solution to the Riemann problem is not computationally efficient as it involves billions of iterations for any practical problem of interest [40]. Approximate, non- iterative solutions have the potential to provide the necessary information for numerical purposes. These approximations get closer to the exact solution as the order is increased, however higher order approximations are not trivial to implement in a software. Hence, SPHinXsys uses a first order approximation called Roe's Riemann solver [48]. This linearized solver has a drawback that it is excessively dissipative [39]. To get around this, SPHinXsys uses a dissipation limiter [39] which is discussed in the next section.

2.2 WCSPH method based on Riemann solver

In SPHinXsys, the WCSPH method based on Riemann solvers is applied for discretizing the continuity and momentum equations as [46,47]

$$\left.\begin{aligned} \frac{d\rho_i}{dt} &= 2\rho_i \sum_j \frac{\frac{m_j}{\rho_j}\left(U^* - \boldsymbol{v_i e_{ij}}\right)\partial W_{ij}}{\partial r_{ij}} \\ \frac{dv_i}{dt} &= -2\sum_j \frac{m_j P^*}{\rho_i \rho_j} \nabla_i W_{ij} \end{aligned}\right\} \quad \textbf{(7)}$$

Here, U^* and P^* are the solutions of an inter-particle Riemann problem along the unit vector $\boldsymbol{e_{ij}} = -\boldsymbol{r_{ij}}/ r_{ij}$ pointing from particle i to j obtained from a low-dissipation Riemann solver [39]. For viscous flows, the physical shear term can be discretized as [34]

$$\left(\frac{dv_i}{dt}\right)^{(\nu)} = 2\sum_j \frac{\frac{\frac{m_j \mu}{\rho_i \rho_j}\boldsymbol{v_{ij}}}{r_{ij}}\partial W_{ij}}{\partial r_{ij}} \quad \textbf{(8)}$$

where μ is the dynamic viscosity.

A first order solution to the Riemann problem is given by Roe's Riemann solver [48] as:

$$\begin{cases} U^* = \bar{U} + \dfrac{\frac{1}{2}(P_L - P_R)}{\bar{\rho} c_0} \\ P^* = \bar{P} + \dfrac{1}{2}\bar{\rho} c_0 (U_L - U_R) \end{cases} \quad \textbf{(9)}$$

where $\bar{U} = (U_L + U_R)/2$ and $\bar{P} = (P_L + P_R)/2$ are interparticle averages. In the SPH code, the subscripts 'L' and 'R' are replaced by 'i' and 'j' respectively.

Since the above discretization is very dissipative, a straightforward modification is to apply a limiter to decrease the implicit numerical dissipation in the intermediate pressure in Eq. (9) as:

$$P^* = \bar{P} + \frac{1}{2}\beta\bar{\rho}c_0(U_L - U_R) \qquad \textbf{(10)}$$

where the limiter β is defined as:

$$\beta = \min\left(\eta_2 \max\left(\frac{U_L - U_R}{c_{LR}}, 0\right), 1.0\right) \qquad \textbf{(11)}$$

β ensures that there is no dissipation when the fluid is under the action of an expansion wave, i.e. $U_L < U_R$, and that the parameter η is used to modulate dissipation when the fluid is under the action of a compression wave, i.e., $U_L < U_R$. Typically, η_2 = 3 is suggested [39].

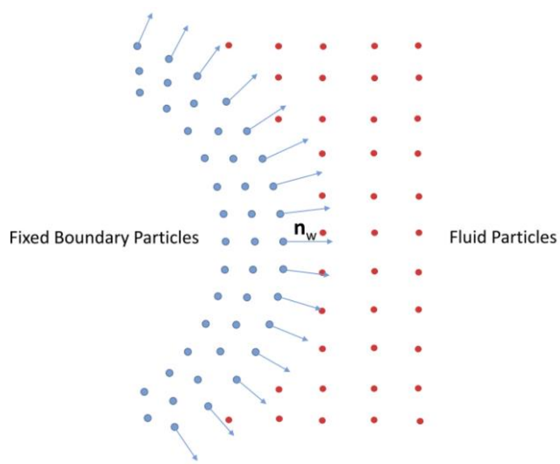


Fig.1. Sketch of fluid particles interacting with fixed wall boundary particles along the wall normal direction through the one-sided Riemann problem (courtesy [39]).

2.3 Wall Boundary condition

Similarly to Adami et al. [49] fixed wall-boundary particles are used, as shown in **Fig.1**, to impose the wall-boundary condition. The interaction between a fluid particle and a wall particle is determined by solving a one-sided Riemann problem [50] along the wall-normal direction. In the one-sided Riemann problem the left state is defined from the fluid particle corresponding to the local boundary normal

$$(\rho_L, U_L, P_L) = (\rho_f, -\boldsymbol{n}_w.\boldsymbol{v}_f, P_f) \qquad \textbf{(12)}$$

where the subscript f represents the fluid particles and $\boldsymbol{n}_w$ the local wall normal direction as shown in **Fig. 1.** According to the physical wall boundary condition, the right state velocity U_R is assumed as

$$U_R = -U_L + 2u_w \qquad \textbf{(13)}$$

where u_w is the wall velocity. The right state pressure is assumed as

$$P_R = P_L + \rho_f \boldsymbol{g}.\boldsymbol{r}_{fw} \tag{14}$$

where $\boldsymbol{r}_{fw} = \boldsymbol{r}_w - \boldsymbol{r}_f$, and the right state density is obtained from eqn. (2). The wall normal direction is calculated for each wall particle using:

$$\boldsymbol{n_w} = \frac{\boldsymbol{\Phi}(\boldsymbol{r_i})}{|\boldsymbol{\Phi}(\boldsymbol{r_i})|}, \qquad \boldsymbol{\Phi}(\boldsymbol{r_i}) = -\Sigma\left(\frac{m_j}{\rho_j}\right)\nabla_{\mathrm{i}} W_{ij} \tag{15}$$

where the summation is over the wall particles only. For problems with static walls, $\boldsymbol{n_w}$ is pre-computed before the numerical solution.

2.4 Surface tension model

With the SPH method, there are generally two ways to model the surface-tension effect: one is based on microscopic inter-phase attractive potentials [51,52]; the other one is based on a macroscopic surface-tension model [53,34]. The first approach is a multiphase one and gives an accurate estimation of the effects of surface tension but involves rather complex calculations of front curvatures that, in some cases, may lead to significant errors [52]. The implementation of the second approach is straightforward and works for single phase as well [52], however, one of the difficulties is that the resulting surface tension needs to be calibrated, i.e., it still has a fudging parameter which is case-dependent [54].

SPHinXsys uses the first approach which I also use for my preliminary multiphase simulations. A detailed description of this approach is beyond the scope of this report. However, the formulation of a surface tension model based on the second approach [55], is given below:

$$F_{ij}^{I} = \frac{\frac{\sigma_s h}{s_2}\boldsymbol{r_{ij}}}{r_{ij}}\begin{cases} \left(s_{ij} - s_1\right)^2 - s_2, & 0 \le s_{ij} < 1 + \frac{s_1}{2} \\ \left(s_{ij} - s_1\right)^2, & 1 + \frac{s_1}{2} \le s_{ij} < 2 \\ 0, & otherwise \end{cases} \tag{16}$$

where F_{ij}^{I} is the inter-particle force between particles i and j, $s_{ij} = \frac{2r_{ij}}{kh}$ and $s_2 = (2 - s_1)^2/2$. s_1 is a fudging parameter controlling the sign of this force and takes values in the range 1.2-1.4 [56]. The force is repulsive when particle spacing satisfies sab < 1 + s1/2, and attractive when particle spacing increases. σ_s is the strength of the force between particles i and j. The maximum attractive force between two particles from Eq. (16) is equal to σ_sh. By the definition of the surface tension coefficient σ, the surface tension force between two particles is on the order of σh. Therefore, σ_sh should also be on the order of σh. For simplicity, σ_s is assumed to have the value of the physical surface tension coefficient σ of the liquid [55].

2.5 Dual-criteria time stepping

SPHinXsys applies the dual-criteria time stepping for the integration of fluid equations. Following Ref. [57], two time-step size criteria, viz. the advection criterion Δt_{ad} is defined as

$$\Delta t_{ad} = CFL_{ad} \min\left(\frac{h}{|v|_{max}}, \frac{h^2}{\nu}\right) \tag{17}$$

and the acoustic criterion Δt_{ac} is given by:

$$\Delta t_{ac} = CFL_{ac} \min\left(\frac{h}{c + |v|_{max}}\right) \tag{18}$$

Here, CFL_{ad} = 0.25, CFL_{ac} = 0.6, $|v|_{max}$ is the maximum anticipated particle advection velocity in the flow and ν denotes the kinematic viscosity. Note that, the calculation of $|v|_{max}$ may consider the external potentials, such as gravity, into consideration. Accordingly, the advection criterion controls the frequency for updating neighboring particles and the acoustic criterion determines the frequency for updating fluid pressure and density.

3. Preliminary experiments

1. 3D droplet impact on solid wall (with surrounding air): In **Fig. 2**, the diameter of the droplet (in blue) was 2.71 mm, and its impact speed was 2.64 m/s. Taking properties of water at STP, this gave a Reynolds number of 7154 and Weber number of 259. A cubic box (in red outline) of air particles side of which is 4 times the droplet diameter is taken. Solid wall particles make up the region between the inner and outer box.

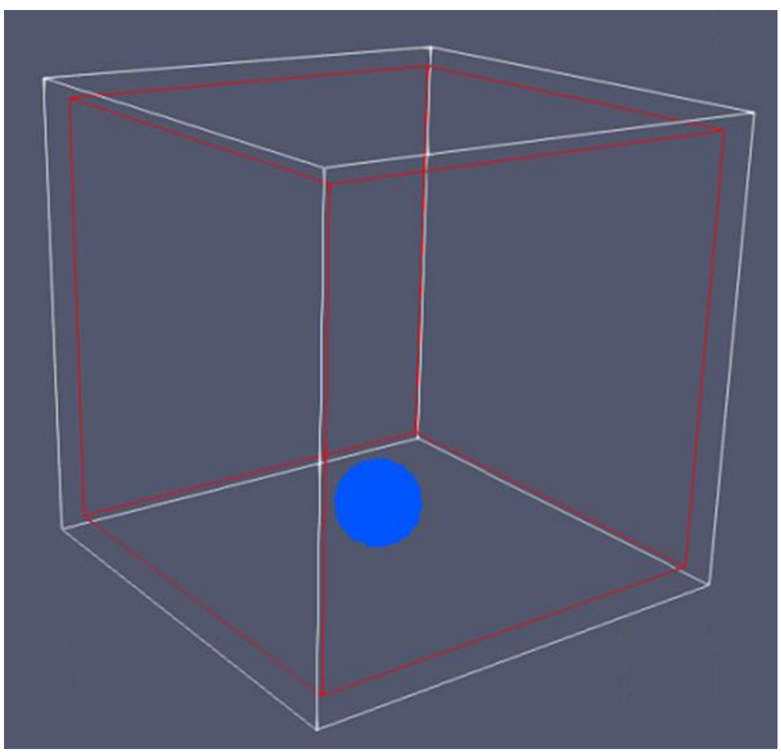


Fig.2 – Water droplet surrounded by air, impacting a solid wall.

The default macroscopic surface tension model in SPHinXsys was chosen for the air-water interaction. For numerical dissipation, a value of 3 was chosen for the fudging parameter η_2 in **Eqn. 11.**

After running the simulation for 4 hours, we reached a physical time of 0.5 ms. **Fig.3** shows a comparison of the shape of the water droplet from our simulation and that from Yang et.al [55].

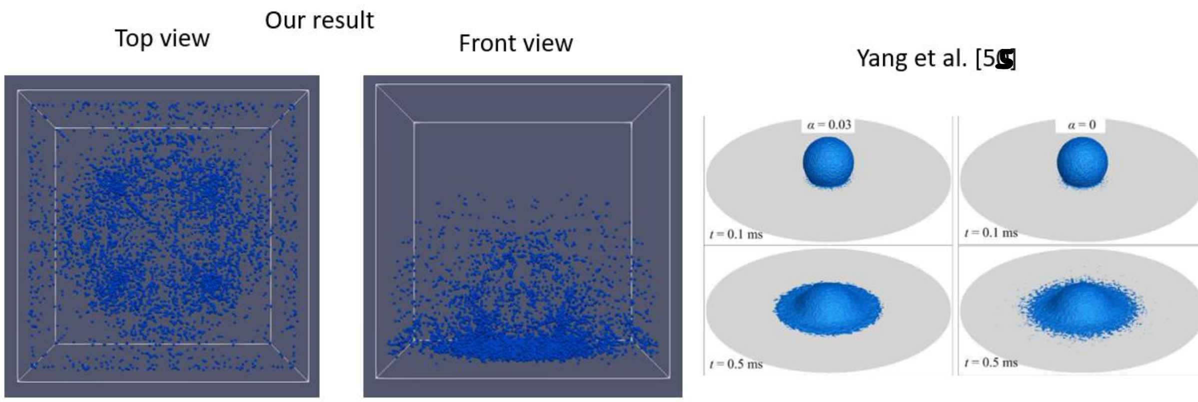


Fig.3. Comparison of our results at 0.5 ms with Yang et. al [55].

In our case, the droplet breaks up completely, instead of only spreading. It's possible that the current surface tension model in SPHinXsys is not robust enough at high Re (>100). To ascertain this, we do some experiments.

2. Square droplet deformation (with surrounding air): In the following three experiments, we investigate the surface-tension driven deformation of an initially square droplet. As shown in **Fig.4(a)**, we place a square patch of water (in blue) into a box of air (in red). In equilibrium, a circular droplet is expected to form, and particles would be at rest (**Fig. 4(b)**).

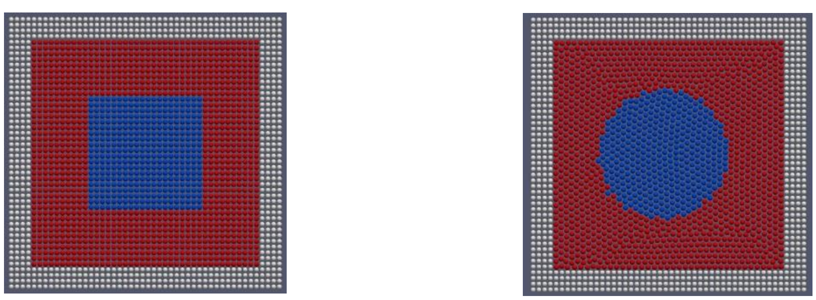


Fig.4. (a) Initial particle positions of the two fluids in a square domain. (b) Circular shape of water droplet in equilibrium.

(a) At low Re: The density and dynamic viscosity of water are taken as 1 and 0.2, and that of air are 0.001 and 0.002 respectively (SI units). The density ratio is 1000 and the viscosity ratio is 100. With a side length of 2 and a reference speed of 5, the Re for the square water patch is 1.625. The surface-tension coefficient is set to 1 and the parameter η_2 is set to 3.

Fig.5 shows the shape of the water droplet in equilibrium. As expected, it is a perfect circle, which proves the efficacy of the multiphase surface tension model at low Re.

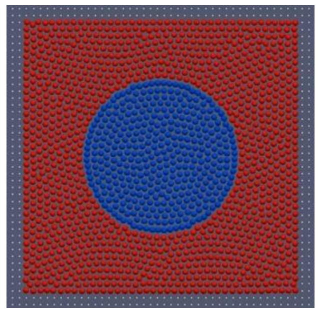
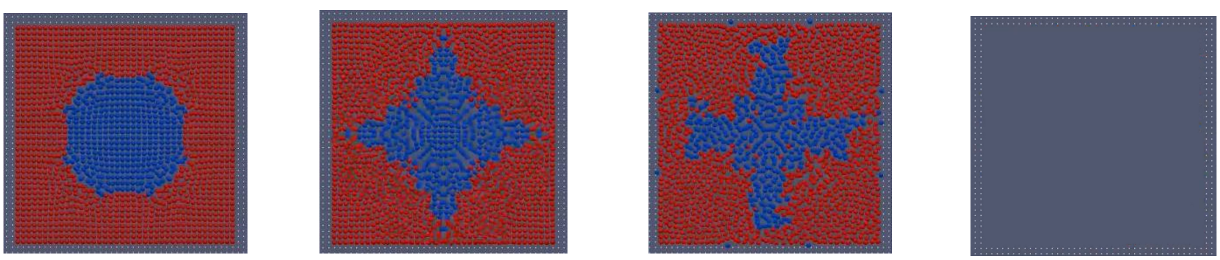


Fig.5. Circular shape of water droplet in equilibrium at low Re.

(b) At high Re: The density and dynamic viscosity of water are taken as 1000 and 0.2, and that of air are 1 and 0.002 respectively (SI units). Note that a density ratio of 1000 and a viscosity ratio of 100 is still maintained. With a side length of 2 and a reference speed of 5, the Re for the square water patch is 1625. The surface-tension coefficient is set to 1 and the parameter η_2 is set to 3.



(a) **(b)** **(c)** **(d)**

Fig.6. Shape of water droplet at Re = 1625 – **(a)** t = 0.1 **(b)** t = 0.2 **(c)** t = 0.3 **(d)** t = 0.4

Fig.6 shows the shape of the water droplet as time progresses. The droplet cannot be stabilized when it has high inertia. In fact, at t = 0.4 (**Fig.6(d)**), all fluid particles disappear due to wrong mathematical treatment of interface curvatures at high Re.

At this point, two possible reasons come to mind; either the viscosity is not high enough or the surface tension model is not robust. To ascertain this, we performed some more experiments.

(c) At high Re (with dissipative Riemann solver): From Eqns. 10 and 11, Roe's Riemann solver is the most dissipative when $\beta = 1$. In literature, it is called the dissipative Riemann solver. Using this value in experiment 2(b), we wanted to see if having maximum numerical dissipation could make the surface tension model work.

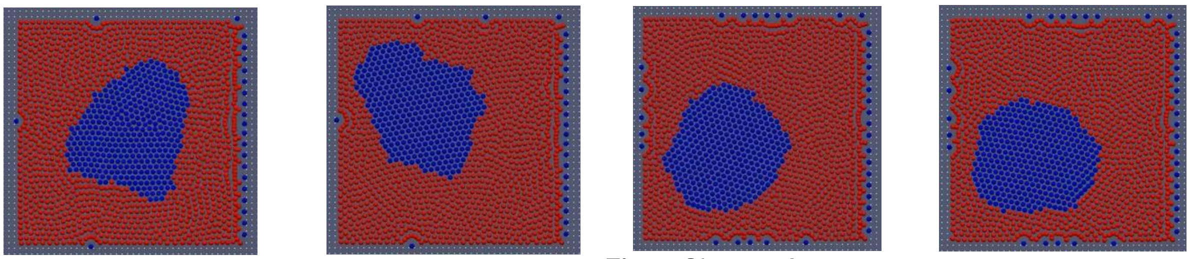


Fig.7. Shape of water droplet at Re = 1625 (with dissipative Riemann solver) – **(a)** t = 0.1 **(b)** t = 0.2 **(c)** t = 0.3 **(d)** t = 0.4

Fig.7 shows the shape of the water droplet as time progresses. The droplet eventually comes to equilibrium at t = 0.4 (**Fig.7(d)**) and assumes a somewhat circular shape. However, it is not at the center of the domain anymore and some water particles have moved through the air phase into the solid wall. This is not physical and is due to wrong mathematical treatment of interface curvatures at high Re. So, at least for the square droplet problem, the surface tension model fails at high Re.

In square droplet example, the water particles are initially at rest. What happens if they are given a velocity to start with? Can the surface tension stabilize the particles? To answer these questions, we carry out a simple experiment of a droplet oscillating after an initial disturbance.

3. Droplet oscillation at low Re: A dynamic test case is the circular liquid-droplet oscillation under the action of capillary forces. Instead of starting from an initially elliptic droplet we prescribe an initial velocity field,

$$\begin{cases} U_x = \frac{U_0 x}{r_0}\left(1 - \frac{y^2}{r_0 r}\right)\exp\left(-\frac{r}{r_0}\right) \\ U_y = \frac{-U_0 y}{r_0}\left(1 - \frac{x^2}{r_0 r}\right)\exp\left(-\frac{r}{r_0}\right) \end{cases} \quad \textbf{(18)}$$

with U_0 = 1 and r_0 = 0.05 for the particles within the drop of radius R = 0.2. r is the distance of the position (x,y) from the droplet center. The computational domain (**Fig.8**) is a box of size 1, and the droplet is initially at the center of the box. The densities of the air phase and the droplet are set to 0.001 and 1, the dynamic viscosities are set to 0.0005 and 0.05 respectively and the surface-tension coefficient between the two phases is 1. At the boundaries we apply no-slip wall boundary conditions. With a reference speed of 1, the Re for this problem is 20.

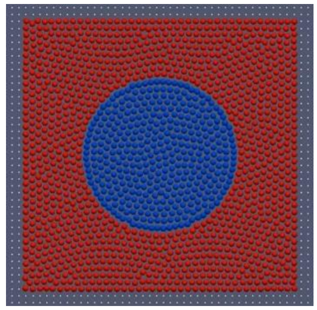


Fig.8. Initial particle setup for the droplet oscillation problem (water in blue, air in red).

In **Fig.9**, we compare the position of the center of mass of the particles in the upper right-quarter section of the droplet with that of Adami et. al [54].

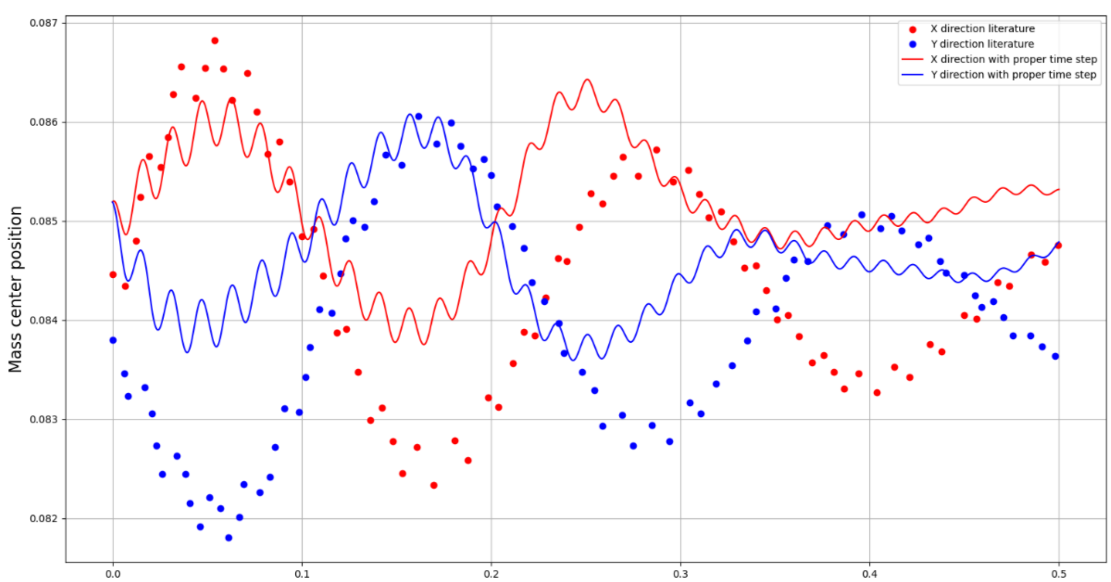


Fig.9. Droplet oscillation with density ratio = 1000: mass center position

For an initial velocity field, even at low Re, the surface tension model in SPHinXsys needs further development so it can capture curvatures of fast-moving interfaces accurately. In SPH, the conclusions drawn from one numerical experiment may not be applicable to a different experiment. Hence, to ascertain the drawback of the surface tension model for our problem of interest, we performed a 2D version of experiment 1, but this time with the dissipative Riemann solver.

4. 2D droplet impact at high Re (with dissipative Riemann solver):

Fig.10 shows the droplet at t = 0 and its subsequent post-impact shapes at t = 0.1, 0.2, 0.5 s.

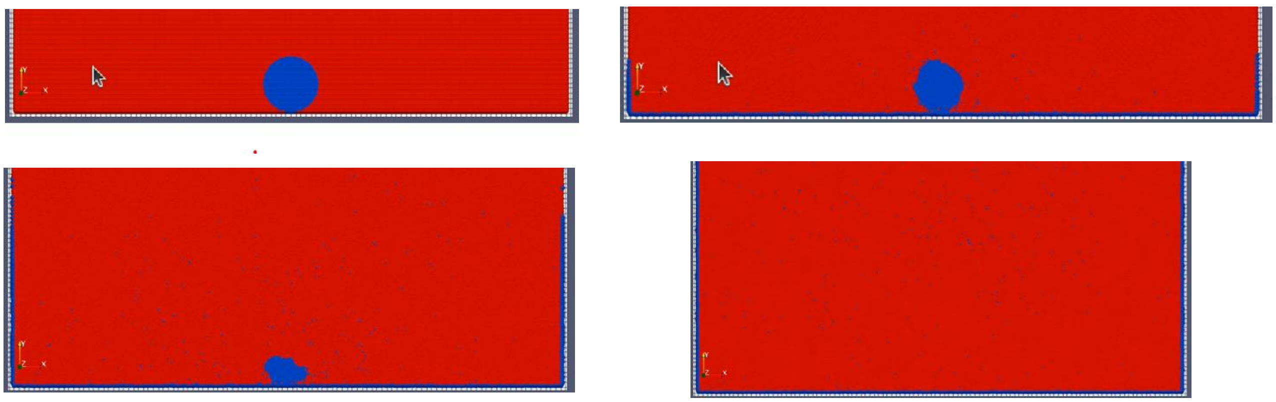


Fig.10. Droplet impact at Re = 7155, We = 260: (a) t=0 (b) t=0.1 (c) t=0.2 (d) t=0.5 s

Although the dissipative Riemann solver is used, the droplet disintegrates completely at t = 0.5 s, with a fraction of particles splashing into the surrounding air and the rest completely spreading on the wall. In fact, the wall is not wide enough to contain the water particles despite the width being 20 times the droplet radius; they move along the side walls too.

Thus, we've ascertained that the current multiphase surface tension model in SPHinXsys is not suitable for a high Re droplet impact problem. This leaves us with the single-phase model in **Eqn.16**. So, we implemented this model in SPHinXsys to simulate the droplet impact problem.

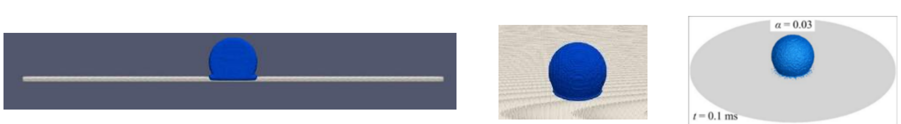
4. Results and discussion

Fig.11 shows the post-impact shape of the droplet in 2D and 3D and compares them against the 3D impact results in Yang et.al [55]. Note that the authors in [55] used artificial viscosity while we used the Riemann solver in Eqns. 9, 10, 11 with the parameter $\eta_2 = 3$. The droplet behaves much better in 2D than in 3D, which is characterized by an unexpected splash at t = 0.5 ms and t = 3.0 ms. The spreading diameter and time are expressed in the following dimensionless form:

$$d^* = \frac{d}{D}, \quad t^* = \frac{tV}{D}$$

where d is the instantaneous spreading diameter and t is the time from impact. d^* is called the spread factor [55]. To get a quantitative estimate of the spread, we plot the spread factor against the non-dimensional time for the 3D droplet impact in **Fig.12**.

(a)



(b)

(c)

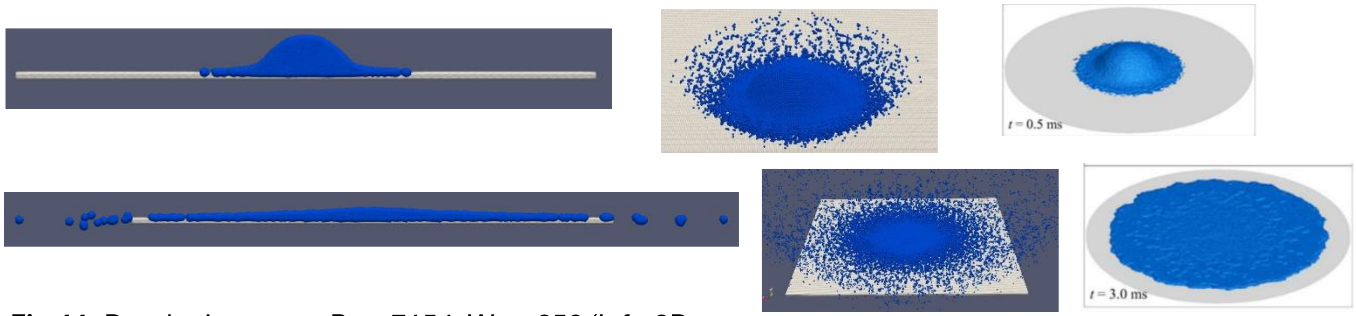


Fig.11. Droplet impact at Re = 7154, We = 259 (left: 2D, center: 3D, right: results from [55]) at (a) t=0.1 ms (b) t=0.5 ms (d) t=3.0 ms

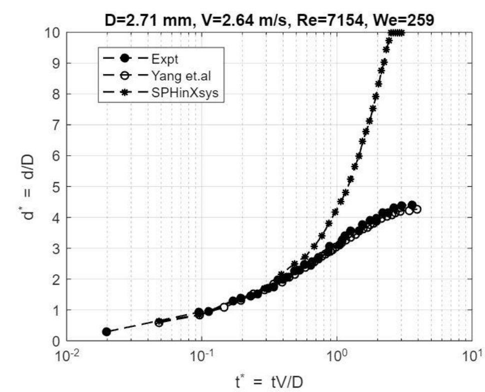


Fig.12. Numerical and experimental [58] results of water drop impact: D = 2.71 mm, V = 2.64 m/s, Re = 7,154, We = 259.

The divergence of the spread ratio in our simulation only reflects the splash of the water droplet in Figures 11(b) and (c). This can be attributed to the low numerical dissipation produced by the Riemann solver when $\eta_2 = 3$. Hence, we checked if the dissipative Riemann solver was able to address this problem, the results of which are shown in **Fig.13**.

(a) (b)

Fig.13. Spreading water droplet post-impact (with dissipative Riemann solver): D = 2.71 mm, V = 2.64 m/s, Re = 7,154, We = 259.

This time, there is no splash and the droplet spreading behavior is closer to Yang et.al [55]. On a more quantitative note, the maximum spread factors are 5.37 and 3.74 in 2D and 3D respectively, while the experimental spread factor in 3D is 4.5 (**Fig.12**). This has two implications: one, that the surface tension force is larger in 3D than in 2D, and second, this Riemann solver is too dissipative to reproduce real physical behavior. The second implication together with **Fig.12** tells us that it is possible to get to the experimental trend of the spread factor using a higher value of η_2. This is like tuning the parameter α in artificial viscosity (Eqn. 6) to match with experiments and possibly has no physical explanation as to why only a particular value would be chosen. This is because the Riemann solver based SPH produces numerical dissipation implicitly, which makes it difficult to quantify the dissipation against the fudging parameter η_2 — current research suggests that only its final effect on the physics can be estimated. This is discussed in a little bit more detail in the Future Work section.

5. Future work

At the time of writing this report, I couldn't find any paper that has discussed a quantitative estimation of the amount of numerical dissipation produced by η_2. This is the need of the hour, simple yet critical enough to convince the community of the credibility of the SPH method as well as the software, SPHinXsys. Recently, an equivalent relation between dissipation limiter η_2 and the artificial viscosity parameter α was theoretically derived by Meng et.al [59]. This is shown below:

$$\eta_2 = \frac{\alpha h(c_i + c_j)}{2(U_L - U_R)|\boldsymbol{r_{ij}}|} \qquad \textbf{(18)}$$

Thus, η_2 can be explicitly programmed as a function of α and can be obtained if α is already known. With this knowledge, it would be useful to find out how much numerical dissipation α generates quantitatively, so it could be traced back to η_2.

Some work has been done in this area where the local effect of artificial viscosity was evaluated directly from the creation of specific entropy for each SPH particle, specific to the problem of rotating discs [60]. Similarly, a method was proposed to reduce numerical artifacts produced by artificial viscosity, for a solid body impacting on a wall type of problem [61]. However, this effect varies from one physical problem to another, with no universal formulation. So, quantifying the effect of α or η_2 for an impact problem is more practical, than doing it for, say Poiseuille flow. Since usage of Riemann solvers in SPH is relatively new, this remains an open research area.

The problem with η_2 came up while trying to reproduce results from Yang et.al [55], where droplet impact without phase change is discussed. For a phase change problem, however, Habashi et.al [62] used a macroscopic and multiphase surface tension model like that in SPHinXsys. Since the SPHinXsys model cannot yet calculate curvatures of fast-moving interfaces accurately, there is room for development in this area too.

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